

Where does space begin?

Earth is cloaked in a thin layer of gases – the atmosphere. Outside this atmosphere is space, where there is no air to breathe, or to allow wings to fly, and where nobody can hear you scream.

Fading away

Our atmosphere does not just end suddenly – it fades gradually into space.

View from *Mir*

Photographed from the American shuttle *Atlantis*, the Russian *Mir* space station once orbited above Earth's atmosphere.



EXOSPHERE



The **exosphere** is the outer layer of the atmosphere, extending about 10,000 km (6,000 miles) above the ground. From here, lighter gases drift into space beyond.

THERMOSPHERE



The **thermosphere** reaches way up to more than 700 km (over 400 miles) above Earth. The polar lights (*aurora borealis* in the north and *aurora australis* in the south) glow in the thermosphere.

MESOSPHERE



Most experts agree that space begins at 100 km (62 miles) above the ground. Past this, our image is not drawn to scale.

STRATOSPHERE



In turn, the **mesosphere** extends about 85 km (53 miles) above the ground. The air is thin here, but it's still thick enough to slow meteorites down.

TROPOSPHERE



The **stratosphere** rises about 50 km (31 miles) above the Earth. Planes cruise in the upper troposphere or lower stratosphere, above the clouds.

The **troposphere** extends between 6 and 20 km (3½–12 miles) above the ground. All our weather takes place in the troposphere.